

Nikita Peganov

AI Researcher

ML researcher working on large language models, multi-agent systems and reinforcement learning, backed by three years of production Python engineering at Yandex. Author of several research works in neuromorphic computing, RL and LLM systems. Currently exploring GenAI security and AI applications in psychology.

Research

LLM Code Translation, Python → Go Yandex, 2026
Applied research, ML Engineer

- built an LLM-based translator porting production Python services to Go
- designed evaluation pipeline for functional correctness of generated code

Multi-Agent LLM Systems 2024–2025
Independent research

- benchmarked open LLMs playing Mafia against each other (local A100 inference)
- built a multi-agent LLM strategy game to study emergent negotiation

RL for Cloud Resource Allocation Huawei, 2021–2022
Researcher, Data Scientist

- trained an RL agent processing 4,500 requests/hour, 1.5M requests in total
- validated applicability of reinforcement learning under real production load

Synchronization of Small-World Neuromorphic Networks RAAI School, 2021–2022
Researcher

- modeled brain-like small-world neuromorphic networks (Kuramoto model)
- studied synchronization dynamics under progressive neuron removal

Selected Projects

EmpathyAI — AI Psychologist Chatbot 2025
Founder, Developer

- AI psychologist chatbot with **100+ active users**; proactive check-ins and long-term analytics
- 2nd round of the “Student Startup” grant (Innovation Promotion Foundation)

Scribo — LLM Thesis Assistant 2024
Founder, Developer

- LLM tool helping students draft course and diploma works; **1,583 users**

frAId — AI Supervisor for Psychotherapists 2023
Researcher, Developer

- CBT supervision bot analyzing sessions and giving feedback to psychologists

Experience

Yandex Feb 2023 — Present
Python Developer, Infrastructure

- develop highload services and metrics collectors for L3 infrastructure management
- built time-series forecasting models predicting service load
- designed an allocator distributing services across balancers via constrained optimization (system of inequalities, cvxpy)

Education

HSE University — MSc, AI Research & Entrepreneurship — GPA 7.38/10 2024–2026

HSE University — BSc, Software Engineering — GPA 6.29/10 2020–2024

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Research Interests

GenAI security (LLM-generated malware) — paper in progress

LLM agents & multi-agent systems

Reinforcement learning

AI in psychology

Skills

Python

PyTorch, NumPy, pandas, scikit-learn, Jupyter

LLMs & NLP

fine-tuning, RAG, prompt engineering, evaluation, local inference

Deep Learning

Transformers, neural networks, generative models

Reinforcement Learning

policy optimization, reward design

Multi-Agent Systems

LLM agents, simulation

Applied ML

time-series forecasting, convex optimization (cvxpy)

Infrastructure

Git, Linux, Docker, CUDA / A100

Languages

English — C1

Russian — Native

Achievements

Research works rated 9/10 by scientific supervisors

Winner, ITMO Open Olympiad in Computer Science (2020)

Winner, United Mathematical Olympiad (2020)